

Digital Signal Intelligence (DSI)

A New Information Substrate for Insurance

THE PROBLEM REINSURANCE HAS ALWAYS HAD

In 2008 global financial crisis shone a light on the consequences of bundled assets whose underlying quality could not be independently verified. Mortgage-backed securities were rated and traded on the assumption that the underlying composition was as described. The opacity was a structural feature of a market that had no independent verification mechanism. Reinsurance has always had the same problem. A treaty reinsurer prices a bundle of risks whose underlying composition it cannot independently inspect. It receives self-reported data and prices on that. The information asymmetry is not exceptional; it is the defining structural feature of every treaty ever written. The underlying portfolio has always been, to a meaningful degree, unknowable.

DSI MAKES THE UNDERLYING PORTFOLIO KNOWABLE

When the direct market operates on DSI — when syndicates, MGAs, and carriers are scoring every risk against observable signals and assigning consistent, verifiable tier classifications — the underlying assets in a reinsurance treaty carry transparent quality scores for the first time. The bundle becomes inspectable. Aggregate exposure can be computed bottom-up from known tier distributions rather than estimated top-down from assumed quality. And entirely new classes of reinsurance product — structured around verified tier tranches, priced on known composition — become possible for the first time in the history of the market

THREE IMPLICATIONS FOR REINSURANCE

Treaty Pricing Transformed

Today, treaty pricing is a function of assumed portfolio quality, moderated by loss history, cedant reputation, and actuarial judgment. When underlying risks carry DSI tier scores, pricing becomes a function of known portfolio quality. A treaty whose underlying book is demonstrably 70% Tier 1–2 by independent DSI assessment is a categorically different risk from one whose cedant claims the same composition. The information asymmetry that has always existed between cedant and reinsurer is resolved.

Tranche Structured Products

Consistent, verifiable tier scoring enables reinsurance products structured around DSI tier tranches — products the market has never been able to offer because underlying quality has never been independently knowable. A Tier 1–2 excess of loss facility priced on verified preferred-risk composition. A Tier 3 aggregate treaty with known elevated-risk characteristics. Risk transfer products whose composition is specified, verified, and auditable, the fixed income market's structured product architecture, applied to insurance risk for the first time.

Cedant Due Diligence

A reinsurance platform running DSI can independently sample and score a cedant's underlying book before treaty attachment, verifying tier distribution claims with observable signal data rather than accepting loss run characterisations. The platform that can say "we independently assessed 100 risks from your book and your stated tier distribution is accurate" is in a fundamentally different negotiating position from one relying on the cedant's own representation. DSI gives reinsurers the independent verification capability the market has never had.

WHAT DSI ENABLES FOR REINSURANCE TODAY

Independent Assessment

DSI scores any entity against observable signals across seven categories without cedant cooperation. A reinsurance platform can assess the underlying quality of any named risk in a treaty portfolio independently, not as a replacement for the cedant's assessment, but as an independent verification of it. The platform that can validate or challenge cedant representations with external signal data occupies a structurally superior negotiating position at every treaty renewal.

Portfolio Aggregation

DSI's organisational graph runtime uses PageRank propagation to surface hidden concentration risk across a reinsured portfolio. Supply chain relationships, shared technology infrastructure, governance network overlaps, the connections that create correlated loss potential across apparently independent risks become visible through multi-hop graph analysis. Accumulation risk that is currently invisible in treaty portfolios becomes measurable and manageable on a common DSI framework.

Continuous Monitoring

DSI is not a point-in-time assessment, it is a continuous monitoring capability with TTL-aware signal refresh. A reinsurance platform can monitor the ongoing digital quality trajectory of treaty portfolios throughout the policy period, generating alerts when monitored risks deteriorate materially between assessments. The platform learns about risk deterioration before loss crystallisation, not at renewal when it is too late to act, and not at claim time when the information serves no pricing purpose.

Facultative Efficiency

For facultative reinsurance, DSI provides independent risk scoring on individual named risks, applying the same seven signal types and tier framework used across the direct market. A facultative submission that arrives pre-scored on DSI from a direct market participant operating on the same standard can be processed at a fraction of current cost and turnaround time. As DSI adoption in the direct market grows; the facultative workflow transforms from bespoke assessment to standard validation.

THE FORWARD POSITION

Reinsurance platforms that deploy DSI now are not merely gaining an efficiency tool for current treaty workflows. They are positioning themselves to define and lead the next generation of reinsurance product architecture, tranche structured, tier-verified, bottom-up aggregated risk transfer products that are only possible when underlying quality is independently scoreable. The platforms that understand this transition earliest will write the products that attract the new capital flows that transparent risk scoring makes possible. The platforms that wait will price the residual book that the structured market leaves behind.

QUESTIONS AND ANSWERS

Q: DSI sounds compelling as a methodology, but what makes it a standard and not another tool?

The traditional underwriting model has existed for over 300 years and has never produced an accepted standard. That is not a coincidence, it is a structural impossibility. A model built on subjective judgment, that is neither canonical nor deterministic, cannot become a standard no matter how long it operates.

DSI has the opposite characteristics. It is observable — every signal is externally verifiable. It is repeatable — the same entity scored twice produces the same result. It is consistent — the algorithm does not vary by individual or institution. It is auditable — every pricing decision traces back through a defined signal-to-tier-to-premium chain. It is non-gameable — signals are drawn from behaviour, not self-reporting. These are precisely the characteristics that allow FICO, PageRank, and credit ratings to function as standards. The traditional model never had them. DSI does.

The path to market acceptance follows a 12–24-month forward validation period — DSI scored risks tracked against actual loss outcomes across early adopters. Once that evidence base exists, the dynamic is self sustaining: capital excluded from insurance by its opacity enters specifically, to back transparently-scored risk.

For reinsurance platforms, the standard argument has a specific additional force: tranche-structured reinsurance products are only viable if the underlying tier scores are trusted by both sides of the transaction. DSI becoming the standard is not just strategically desirable for this audience, it is the precondition for the new product architecture to exist at all.

Q: The MBS parallel is provocative. Is the reinsurance opacity problem really that serious?

The parallel is precise, not provocative. The structural feature that caused 2008 — pricing bundles of assets whose underlying quality could not be independently verified at scale — is present in every reinsurance treaty ever written. The consequences in insurance have been less catastrophic because insurance losses are less correlated than mortgage defaults, not because the information asymmetry is smaller. Reinsurance pricing has always incorporated an implicit uncertainty premium to compensate for unknowable underlying composition. DSI does not eliminate that uncertainty — it replaces it with measurable, verifiable quality data. The premium that was compensation for opacity becomes available as margin improvement when opacity is removed.

Q: How does tranche-structured reinsurance work in practice with DSI?

The mechanism is straightforward once underlying risks carry consistent DSI tier scores. A cedant operating on DSI can present a treaty submission that specifies not just premium volume and loss history, but verified tier distribution — "this book is 68% Tier 1–2, 24% Tier 3, 8% Tier 4–5 by independent DSI assessment." The reinsurance platform can independently verify that distribution by sampling the underlying risks. Treaty terms, attachment points, and pricing are then negotiated on the basis of known composition rather than assumed quality. Over time, standardised tranche products emerge — Tier 1–2 preferred facilities, Tier 3 elevated-risk aggregates — with pricing that reflects verified underlying quality rather than actuarial estimates of unknown composition.

Q: How does DSI handle the correlation risk that is specific to reinsurance portfolios?

DSI's organisational graph runtime builds relationship graphs with six node types and six edge types, then applies PageRank propagation to derive network authority scores through multiple relationship hops. This multi-hop analysis surfaces the shared infrastructure, supply chain dependencies, and governance network overlaps that create correlated loss potential across apparently independent risks. For a reinsurance portfolio, this means accumulation risk that is invisible in individual risk assessments — and entirely absent from traditional treaty analysis — becomes computable from the bottom up. The platform can quantify concentration to a sector, a technology dependency, or a counterparty network across the entire treaty book simultaneously..